Mohamed El Alami

Computer Science · AI Safety Research (prospective)

Shenzhen, China elalamimeded@gmail.com github.com/elalamiimed hitszcs.com

RESEARCH INTERESTS

AI safety, focused on the evaluation and oversight of agentic systems: how LLM agents are tested for long-horizon failure modes, how memory and tool use change behaviour, what interpretability can reveal about internal state, and how explicit specifications — guardrails, rejection checklists, fail-closed gates — can be made testable. Background in empirical machine learning and systems engineering. Seeking MSc/PhD supervision (2028 intake); open to research assistantships.

EDUCATION

Harbin Institute of Technology, Shenzhen

2024 – 2028 (expected)

B.Eng. Computer Science & Technology

Coursework: machine learning (SVM, ensembles, PCA, neural networks, ROC/AUC), data structures & algorithms, databases and SQL, computer networks, signals & systems, object-oriented programming. Elected class monitor 2024–2025; invited orientation-day speaker.

Qingdao Agricultural University

2023 – 2024

Chinese Language Program — merit certificate (优秀证书)

RESEARCH & ENGINEERING PROJECTS

LearnyBrain — adaptive tutoring agent

private

Agent harness that probes where a learner's understanding ends, maps the dependency path to a goal, then teaches and verifies one reasoning step at a time; a persistent knowledge graph carries state across sessions. An exercise in agent state, step-wise verification and long-horizon coherence.

TypeScript · Next.js · Electron

Amnesia — local-first memory for LLM agents

public

Long-term memory layer for Ollama models: hybrid SQLite full-text and embedding recall behind an MCP interface, no cloud and no telemetry. The research question is the audit surface — what an agent stores, when it retrieves it, and how contradictory memories change behaviour.

Python · SQLite · MIT · ~11.5k lines · 16 test files

Agentica — guarded bilingual writing pipeline

public

A 27-question intake produces a publication-ready EN/中文 draft, which is then gated by a deterministic rejection checklist distilled from 17 approved articles and 130 headings. An exercise in constraining generation with explicit, testable criteria.

Python · LLM APIs · 13 QA test files

CryptoBot — empirical study of 41,470 token launches

public

Labelled launch corpus (41,470 launches, 45,000+ on-chain tapes) used to test whether early observables predict extreme outcomes. A gradient-boosted classifier improved from ROC-AUC 0.60 to 0.75 across launch windows; 18 written reports. Paper trading only — no wallet, no signing.

Python · pandas · scikit-learn · XGBoost

MISO — 20-engine gold macro research system

private

Twenty signal engines (COT positioning, intermarket, volatility, ETF flows, COMEX, USD liquidity) aggregate into weekly and daily decisions, with an 18-tab dashboard and reproducible backtests. Every signal is traceable to its input and every decision reproducible from stored state.

Python · FastAPI · SQLite · Streamlit · 82k lines · 486 files · 37 test files

Pump Terminal — fail-closed trading terminal

local only

Production-oriented Solana terminal with realtime candles and tape, volume profile, deterministic paper replay and a three-gate risk engine. The live gate fails closed: if a precondition cannot be verified, the system does nothing — the closest engineering analogue to enforcing hard constraints.

FastAPI · Next.js · Postgres/TimescaleDB · Redis · 116k lines · 552 Python tests

Session-Sweep Engine — strategy as an evaluable specification

public

Event-driven Backtrader backtester for an XAUUSD session-liquidity-sweep strategy; every rule from the playbook became an explicit, testable precondition (seven gates, 2% risk model, daily circuit breaker) on real gold data.

Python · Backtrader · Pydantic

TECHNICAL SKILLS

LANGUAGES

Python (primary), TypeScript / JavaScript, Java, C++ (coursework), SQL

ML

scikit-learn, pandas, XGBoost, evaluation methodology

SYSTEMS

FastAPI, Flask, Node.js, Linux, Docker, Git

INTERFACES

React, Next.js

TEACHING, MENTORING & SERVICE

HITSZCS — founder & maintainer

2024 — present

Open course-material library for HITSZ computer-science students: 53 stars, 9 forks, 563 commits, 16 contributors, live at hitszcs.com. Ran GitHub workshops and described the project in the HITSZ orientation address.

HSIYC — Director, General Affairs (综合事务部 部长)

2026 — present

Lead a 23-person department in HITSZ's first formally recognised international student organisation (167 members); ran department recruitment (127 applicants) and authored the club's governance and progress documentation.

Hack Hive Hackathon 01 — organiser & judge

2026

Designed a 71-hour hackathon for two-person teams, built the event site and PHP team-matching backend, wrote the rules and judged six submissions.

Class monitor — HITSZ, Class 2 (elected)

2024 — 2025

Chaired class committee meetings and represented classmates, including escalating cohort-wide concerns about a mid-term teaching method.

LANGUAGES

Arabic (native) · English (fluent) · French (fluent) · Chinese 中文 (intermediate) · Spanish (beginner)
References available on request.